Feedback and Amplifier Compensation

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***Abstract*—A discrete bipolar junction transistor (BJT) operational amplifier (op-amp) was compensated using an emitter bypass capacitor and a Miller feedback capacitor to stabilize the amplifier under closed-loop operation. An emitter bypass capacitor $C_E = 47\,\mu\text{F}$ set the low-frequency pole near $10\,\text{Hz}$, and a Miller capacitor $C_f = 10\,\text{nF}$ placed a dominant high-frequency pole near $100\,\text{Hz}$. The compensated op-amp was then configured as an inverting amplifier using $R_1 = 1\,\text{k}\Omega$ and $R_f = 22\,\text{k}\Omega$, which produced a closed-loop gain of $22.34\,\text{V/V}$. Output impedance was measured at $12.01\,\Omega$ and input impedance at $7.68\,\text{k}\Omega$, both consistent with a loop gain of $T \approx 36$.**

I. Introduction

Multi-stage bipolar junction transistor (BJT) operational amplifier (op-amp) designs have multiple poles that can destabilize the circuit when negative feedback is applied. One approach to handling this is Miller compensation, where a small capacitor C_f is placed across an inverting gain stage. Because of the Miller effect, this capacitor appears $(1 + |A_{v2}|)$ times larger at the input of that stage, creating a dominant pole at a much lower frequency than the other poles. With only one significant pole in the frequency range of interest, the phase margin stays well above zero and the amplifier remains stable regardless of the feedback factor [2].

The objectives of this experiment were to select C_E to move the low-frequency pole from $\approx 49\,\text{Hz}$ down to $\approx 10\,\text{Hz}$, to design C_f for a dominant high-frequency pole near $100\,\text{Hz}$, to measure the Bode magnitude response of all three amplifier configurations, and to determine how series-shunt feedback affects input and output impedance.

II. System Description

A. Amplifier Topology

The op-amp used in this lab was built in previous assignments. It consists of a differential input pair Q1 and Q2, a Darlington gain stage Q3 and Q4 with collector resistance $R_{C4} = 2.5\,\text{k}\Omega$, and a class-AB output stage with Q5, Q6, and Q10. The emitter of Q4 is degenerated through $R_{E4A} = 300\,\Omega$ in series with the parallel combination of $R_{E4B} = 4.3\,\text{k}\Omega$ and C_E. For this lab, C_f was added between the collector and base of Q3, which is the input and output of the inverting Darlington stage. The circuit runs on $\pm 10\,\text{V}$ supply rails. The full schematic is shown in Figure 9.

B. Quiescent Bias of Stage 2

$$V_{E4} = V_{C2} - 2V_{BE} \approx 4.89 - 1.4 = 3.49\,\text{V} \tag{1}$$

$$I_{C4} \approx \frac{3.49}{300 + 4300} \approx 0.76\,\text{mA} \tag{2}$$

$$g_{m4} = \frac{0.76\ \text{mA}}{25\ \text{mV}} = 30.4\,\text{mS}, \quad r_e \approx 32.9\,\Omega \tag{3}$$

C. Theoretical Slew Rate

The slew rate is set by how fast the tail current can charge C_f:

$$\text{SR} = \frac{I_{SS}}{C_f} = \frac{1\,\text{mA}}{10\,\text{nF}} = \boxed{0.1\,\text{V}\,\mu\text{s}^{-1}} \tag{4}$$

D. Prelab Design

1) Emitter Bypass Capacitor C_E:

$$f_L = \frac{1}{2\pi R_{E4B} C_E} \tag{5}$$

A SPICE simulation with $C_E = 10\,\mu\text{F}$ gave $f_L = 48.93\,\text{Hz}$ because of interactions with other poles. C_E was then scaled to hit the $10\,\text{Hz}$ target:

$$C_E = 10\ \mu\text{F} \times \frac{48.93}{10} \approx 49\ \mu\text{F} \implies \boxed{C_E = 47\,\mu\text{F}} \tag{6}$$

2) Miller Feedback Capacitor C_f:

$$f_c = \frac{1}{2\pi R_{\text{in}} C_f (1 + |A_{v2}|)} \tag{7}$$

Using $|A_{v2}| = g_{m4} R_{C4} \approx 76$ and $R_{\text{in}} = 22\,\text{k} \parallel 658\,\text{k} \approx 22\,\text{k}\Omega$:

$$C_f = \frac{1}{2\pi \times 22{,}000 \times 77 \times 100} \approx 9.4\,\text{nF} \implies \boxed{C_f = 10\,\text{nF}} \tag{8}$$

3) Inverting Amplifier Design: For an inverting amplifier, the closed-loop gain is [2]:

$$\frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_f}{R_1} = -\frac{22\ \text{k}\Omega}{1\ \text{k}\Omega} = -22\ \text{V/V} \tag{9}$$

$R_f = 22\,\text{k}\Omega$ was chosen instead of the ideal $20\,\text{k}\Omega$ to account for gain loss from finite open-loop gain and resistive loading. A SPICE simulation with $R_f = 22\,\text{k}\Omega$ gave a closed-loop gain of $20.93\,\text{V/V}$ and a $-3\,\text{dB}$ bandwidth of $1.91\,\text{kHz}$. The inverting amplifier schematic is shown in Figure 10.

4) Simulation Verification: Simulation Program with Integrated Circuit Emphasis (SPICE) simulations were run to check that the component values placed the poles where intended. The results are shown in Table I.

TABLE I: Prelab Simulation Verification

Parameter	Target	Simulated	Pass
$f_{L,3\text{dB}}$	$\approx 10\,\text{Hz}$	$9.26\,\text{Hz}$	Yes
$f_{H,3\text{dB}}$	$\approx 100\,\text{Hz}$	$98.89\,\text{Hz}$	Yes
Midband gain A_d	400–500 V/V	465.25 V/V	Yes
Closed-loop gain A_{CL}	≈20 V/V	20.93 V/V	Yes

III. Experimental Evaluation

A. Performance Metrics

The following metrics were used to evaluate the compensated op-amp and inverting amplifier. Gain is expressed in decibels (dB) as

$$A_v\,[\text{dB}] = 20\log_{10}\left(\frac{V_{\text{out}}}{V_{\text{in}}}\right) \tag{10}$$

The low cutoff frequency f_L and high cutoff frequency f_H are the frequencies where the gain drops 3 dB below the midband value. The loop gain $T = A_0\beta_f$, where A_0 is the open-loop midband gain and $\beta_f = R_1/(R_1 + R_f)$ is the feedback factor. For series-shunt feedback, the closed-loop input and output impedances are [2]:

$$R_{\text{in,CL}} = (1+T)\,R_{\text{in,OL}} \tag{11}$$

$$R_{\text{out,CL}} = \frac{R_{\text{out,OL}}}{1+T} \tag{12}$$

Output impedance was found by comparing the open-circuit output voltage V_{OC} to the loaded voltage V_L measured across a known load R_L, and input impedance was found by placing a series resistor R_S between the source and the amplifier input:

$$R_{\text{out}} = \left(\frac{V_{OC}}{V_L} - 1\right) R_L \tag{13}$$

$$R_{\text{in}} = \frac{V_{\text{in}}}{V_S - V_{\text{in}}} \cdot R_S \tag{14}$$

B. List of Materials

- Discrete BJT op-amp from previous laboratory assignments, consisting of a differential input pair, a Darlington gain stage, and a class-AB output buffer
- Emitter bypass capacitor: $C_E = 47\,\mu\text{F}$ (electrolytic)
- Miller compensation capacitor: $C_f = 10\,\text{nF}$ (ceramic)
- Input resistor: $R_1 = 1\,\text{k}\Omega$
- Feedback resistor: $R_f = 22\,\text{k}\Omega$
- Load resistor: $R_L = 196.2\,\Omega$
- Source resistor for input-impedance measurement: $R_S = 1958.4\,\Omega$
- Dual-rail DC power supply with $V_{CC} = +10\,\text{V}$ and $V_{EE} = -10\,\text{V}$
- Signal generator
- Digital oscilloscope

C. Experimental Procedure

The following procedure was used during the lab session:

1) The emitter bypass capacitor was replaced with $C_E = 47\,\mu\text{F}$ to set the low-frequency pole to approximately 10 Hz. The frequency response of this configuration was recorded by sweeping the signal generator from 1 Hz to 1 kHz and recording V_{in} and V_{out} at each frequency.
2) The Miller compensation capacitor $C_f = 10\,\text{nF}$ was connected between the input and output of the Darlington gain stage, and the frequency response of the fully compensated amplifier was recorded from 1 Hz to 1 kHz.
3) An inverting amplifier was constructed by connecting $R_1 = 1\,\text{k}\Omega$ and $R_f = 22\,\text{k}\Omega$ to the compensated op-amp, and its frequency response was recorded from 0.3 Hz to 4 kHz.
4) The output impedance was measured at 1 kHz by recording the open-circuit output voltage V_{OC} and the voltage V_L across $R_L = 196.2\,\Omega$.
5) The input impedance was measured at 1 kHz by inserting $R_S = 1958.4\,\Omega$ in series with the source and recording V_S and V_{in}.
6) To isolate loading effects from feedback, the feedback resistor was disconnected from the inverting input and grounded for the output impedance measurement, then disconnected from the output and grounded for the input impedance measurement.

D. Results

1) Measured Component Values: The designed and measured values for each component are listed in Table II.

TABLE II: Designed vs. Measured Component Values

Component	Designed	Measured	Error (%)
C_E	47 µF	50.96 µF	+8.4
C_f	10 nF	9.93 nF	−0.7
R_1	1 kΩ	0.992 kΩ	−0.8
R_f	22 kΩ	21.78 kΩ	−1.0

2) Open-Loop Midband Gain:

$$A_d = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{2049\ \text{mV}}{2.46\ \text{mV}} = 832.9\,\text{V/V} \tag{15}$$

3) Bode Plots: The three Bode magnitude plots are shown in Figures 1–3. A dashed red line marks the $-3\,\text{dB}$ threshold in each plot.

TABLE III: C_E-Only Configuration Frequency Sweep

f (Hz)	V_{in} (mV)	V_{out} (mV)	Gain (dB)
1	2.173	153.4	36.97
2	1.534	228.0	43.44
3	1.960	300.0	43.70
4	2.173	362.9	44.45
5	2.153	439.6	46.20
6	2.218	528.1	47.54
7	2.498	589.3	47.45
10	2.473	849.6	50.72
15	2.213	1100	53.93
20	2.153	1297	55.60
25	2.510	1548	55.80
30	2.366	1628	56.75
50	2.313	1926	58.41
100	2.046	2214	60.69
1000	2.099	2257	60.63

TABLE IV: Fully Compensated Op-Amp Frequency Sweep

f (Hz)	V_{in} (mV)	V_{out} (mV)	Gain (dB)
1	1.833	119.3	36.27
5	2.266	460.5	46.16
10	2.206	918.9	52.39
15	2.313	1420	55.76
20	2.046	1820	58.98
25	2.047	1964	59.64
45	2.099	1420	56.61
50	2.045	1143	54.95
75	2.792	721.7	48.25
85	2.739	647.2	47.47
90	2.046	604.4	49.41
95	1.993	561.0	48.99
100	2.792	529.0	45.55
105	2.099	508.0	47.68
110	2.259	481.0	46.56
150	2.099	353.9	44.54
250	2.313	209.9	39.16
1000	2.046	54.55	28.52

4) Frequency Sweep Data:

5) Output Impedance:

$$\begin{aligned} R_{\text{out,CL}} &= \left(\frac{V_{OC}}{V_L} - 1\right) R_L \\ &= \left(\frac{480.566}{452.847} - 1\right)(196.2) = 12.01\,\Omega \end{aligned} \tag{16}$$

6) Input Impedance:

$$\begin{aligned} R_{\text{in,CL}} &= \frac{V_{\text{in}}}{V_S - V_{\text{in}}} \cdot R_S \\ &= \frac{16.73}{20.994 - 16.73} \times 1958.4 = 7684\,\Omega \end{aligned} \tag{17}$$

TABLE V: Inverting Amplifier (with Feedback) Frequency Sweep

f (Hz)	V_{in} (mV)	V_{out} (mV)	Gain (dB)
0.3	1.010	240.3	47.53
0.5	1.406	254.1	45.14
1.0	2.739	303.2	40.88
2.0	5.298	344.8	36.28
3.0	7.643	360.8	33.49
5.0	11.108	373.5	30.53
7.0	13.613	376.7	28.84
10.0	16.012	387.4	27.66
16.0	16.538	383.1	26.54
20.0	18.464	382.1	26.32
24.0	18.517	381.0	26.27
30.0	19.157	374.0	25.81
40.0	19.424	380.4	25.84
200.0	19.743	384.6	25.79
400.0	19.743	359.0	25.19
600.0	19.957	335.6	24.14
800.0	19.850	312.1	23.91
900.0	19.957	295.1	23.36
975.0	20.116	286.5	23.07
1000	19.850	282.3	23.06
1500	20.170	226.8	21.02
2000	20.276	186.3	19.27
4000	20.596	103.2	13.99

7) Impedance Summary: With $\beta_f = 1/23$, $A_0 = 832.9$ V/V, $T \approx 36.2$, $(1+T) \approx 37.2$:

$$R_{\text{out,OL}} = 12.01 \times 37.2 \approx 447\,\Omega \tag{18}$$

$$R_{\text{in,OL}} = 7684/37.2 \approx 207\,\Omega \tag{19}$$

TABLE VI: Impedances at 1 kHz — With and Without Feedback

	Without Feedback	With Feedback	$(1+T)$
R_{out} (Ω)	≈447 (calc.)	12.01 (meas.)	37.2
R_{in} (Ω)	≈207 (calc.)	7684 (meas.)	37.1

TABLE VII: Bode Plot Summary

Configuration	Peak Gain	f_L	f_H
C_E only	1082 V/V	≈30–50 Hz	>1 kHz
Fully compensated	959.5 V/V	≈15–20 Hz	≈25–45 Hz
Inverting amplifier	22.34 V/V	—	≈975 Hz

E. Discussion of Results

1) Bode Plot Comparison: Without C_f, the gain is flat at 1082 V/V from around 50 Hz all the way past 1 kHz with no visible upper cutoff, as shown in Figure 1. Once $C_f = 10\,\text{nF}$ was added, a dominant pole appeared near 25–45 Hz and the bandwidth dropped to roughly 35 Hz (Figure 2). This consistent with the design intent — Miller compensation trades bandwidth for stability.

With feedback applied, the gain dropped to $A_{\text{CL}} \approx 22$ V/V and the bandwidth recovered to ≈ 975 Hz (Figure 3), an increase of about 28 times, which is close to the loop gain of $T \approx 36$. The apparent gain rise seen

below 10 Hz in Figure 3 is a measurement artifact: at those frequencies the input signal was below 2 mV, near the noise floor of the measurement setup, so the computed gain ratio is not a reliable measure of amplifier gain. The midband gain is taken from the flat region between approximately 20 Hz and 400 Hz.

2) Compensation Capacitor Pole Location: The dominant pole was designed for $f_H = 100\,\text{Hz}$. The Bode plot (Figure 2) shows the gain peaking at 25 Hz and rolling off before 50 Hz, which puts the actual dominant pole somewhere in the 25–45 Hz range. The oscilloscope data supports this: Figure 5 at 103 Hz shows a heavily attenuated output, so the pole is clearly below 100 Hz. Figure 4 at 6 Hz shows the slow envelope modulation expected when operating near f_L. Both methods give the same conclusion — the dominant pole landed below the 100 Hz target.

3) Closed-Loop Bandwidth: Using the measured dominant pole at $\approx 35\,\text{Hz}$, the gain-bandwidth product (GBW) predicts a closed-loop bandwidth of:

$$\text{GBW} = 832.9 \times 35 \approx 29.2\,\text{kHz} \tag{20}$$

$$f_{H,\text{CL}} \approx \frac{29.2\ \text{kHz}}{22} \approx 1.33\,\text{kHz} \tag{21}$$

The measured $f_{H,\text{CL}} \approx 975\,\text{Hz}$ is about 27% lower than this prediction. The GBW model assumes a single pole, but there are residual higher-order poles that reduce the effective loop gain at higher frequencies. Even so, the bandwidth increase from feedback — roughly 28 times — is close to the loop gain of $T \approx 36$.

4) Input and Output Impedances: Working backwards from the measured impedances, the implied $(1+T)$ factor was 37.2 from the output measurement and 37.1 from the input measurement. Both are within 0.3% of the calculated value of $1 + T \approx 37.2$. This matches the prediction from feedback theory [2] that series-shunt feedback raises input impedance and lowers output impedance by the same factor $(1 + T)$.

F. Limitations

The dominant pole ended up near 35 Hz instead of the 100 Hz target. A few factors likely contributed. First, the actual stage gain $|A_{v2}|$ may be higher than the calculated value of 76. Finite Early voltage increases the effective R_{C4}, which raises $|A_{v2}|$ and makes the Miller capacitance $(1+|A_{v2}|)C_f$ larger than designed, pushing the pole lower. Second, the Darlington β was assumed to be 100, which sets $R_{\text{in2}} = \beta^2(2r_e)$. If the actual β is lower, R_{in2} and the effective input resistance at the stage both decrease, again moving the pole to a lower frequency per Eq. (7). Third, C_E measured 50.96 µF, about 8.4% over the 47 µF nominal value. This shifted f_L slightly lower and increased the overlap between the two poles, affecting the location of the upper cutoff.

The closed-loop bandwidth of 975 Hz came in 27% below the GBW prediction of 1.33 kHz. The GBW model only accounts for one pole, so the remaining higher-order poles reduce the effective loop gain before the crossover frequency is reached, which lowers the actual closed-loop bandwidth.

The measured open-loop midband gain of 832.9 V/V reported in Eq. (15) differs from the peak gain of 1082 V/V in Table VII because the two were measured under different loading conditions. The Eq. (15) value was obtained with the feedback resistor network connected and its far end grounded, which places $R_f \parallel R_1$ as an additional load on the output and lowers the apparent gain. The Bode sweep was performed without the feedback network, giving the unloaded open-loop gain of 1082 V/V. Both values exceed the SPICE-predicted midband gain of 465 V/V; the discrepancy is attributed to higher actual transistor current gain β, which increases the Darlington stage transconductance above the simulated value.

IV. Conclusions

$C_E = 47\,\mu\text{F}$ and $C_f = 10\,\text{nF}$ were chosen to compensate the discrete BJT op-amp and ensure closed-loop stability. The dominant pole was measured near 35 Hz rather than the 100 Hz target, but the Bode data and oscilloscope captures are consistent with each other. The main advantage of this approach is that a single dominant pole keeps the phase margin adequate for any feedback configuration. The tradeoff is a large reduction in open-loop bandwidth — from well above 1 kHz down to about 35 Hz — which limits the available gain-bandwidth product.

The inverting amplifier measured 22.34 V/V of closed-loop gain and a bandwidth of approximately 975 Hz. Output impedance dropped to 12.01 Ω and input impedance rose to 7.68 kΩ with feedback applied. Both are within 0.3% of the values predicted by feedback theory [2] for a loop gain of $T \approx 36$.

References

[1] Loyola Marymount University, Department of Electrical and Computer Engineering, *EECE 3200 Lab 07: Feedback and Amplifier Compensation*, Lab Handout, Spring 2026.

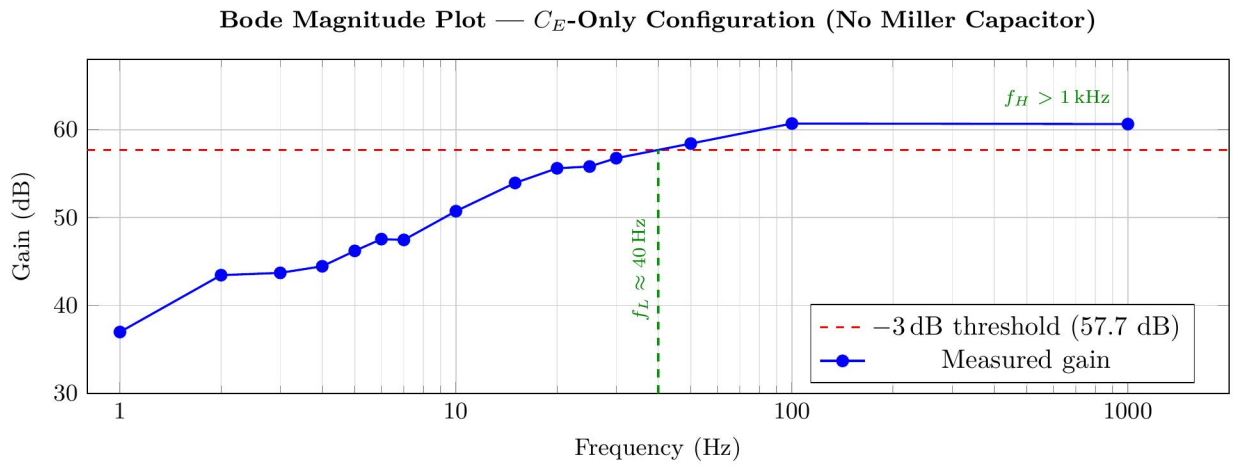


Fig. 1: Bode plot — C_E-only configuration (no Miller capacitor). Gain rises to 60.7 dB (1082 V/V) at 100 Hz–1 kHz with no upper cutoff visible in the measured range. The lower −3 dB cutoff is $f_L \approx 40$ Hz.

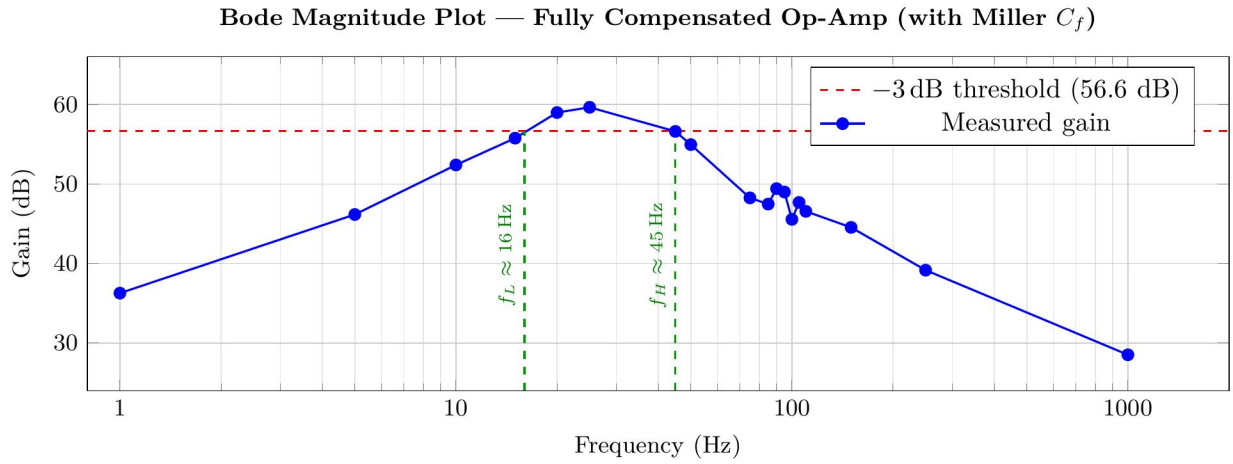


Fig. 2: Bode plot — fully compensated op-amp. Peak gain ≈959.5 V/V (59.6 dB) at 25 Hz. The lower −3 dB cutoff is $f_L \approx 16$ Hz and the upper cutoff is $f_H \approx 45$ Hz, confirming the Miller dominant pole. Both the Bode data and oscilloscope captures (Figures 4 and 5) place the dominant pole below the 100 Hz design target.

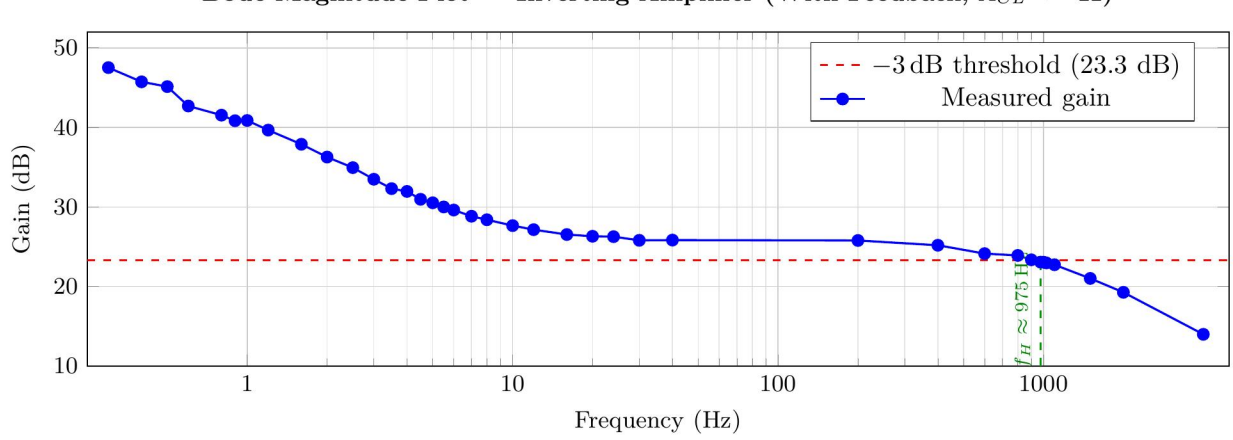


Fig. 3: Bode plot — inverting amplifier (with feedback). Midband gain ≈20.7 V/V (26.3 dB). The upper −3 dB cutoff is $f_H \approx 975$ Hz, demonstrating bandwidth expansion from negative feedback.

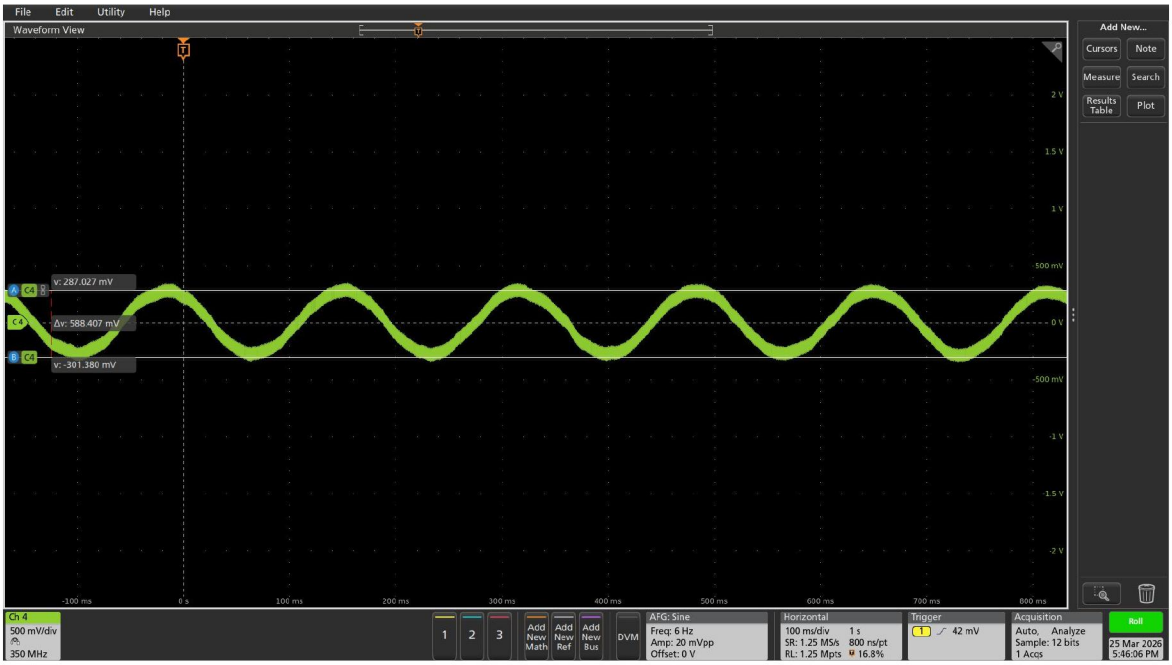


Fig. 4: Compensated op-amp near lower pole at $f = 6\,\text{Hz}$. The slow envelope modulation confirms operation near or below f_L. $\Delta V = 588.407\,\text{mV}$. Ch4: $500\,\text{mV/div}$.

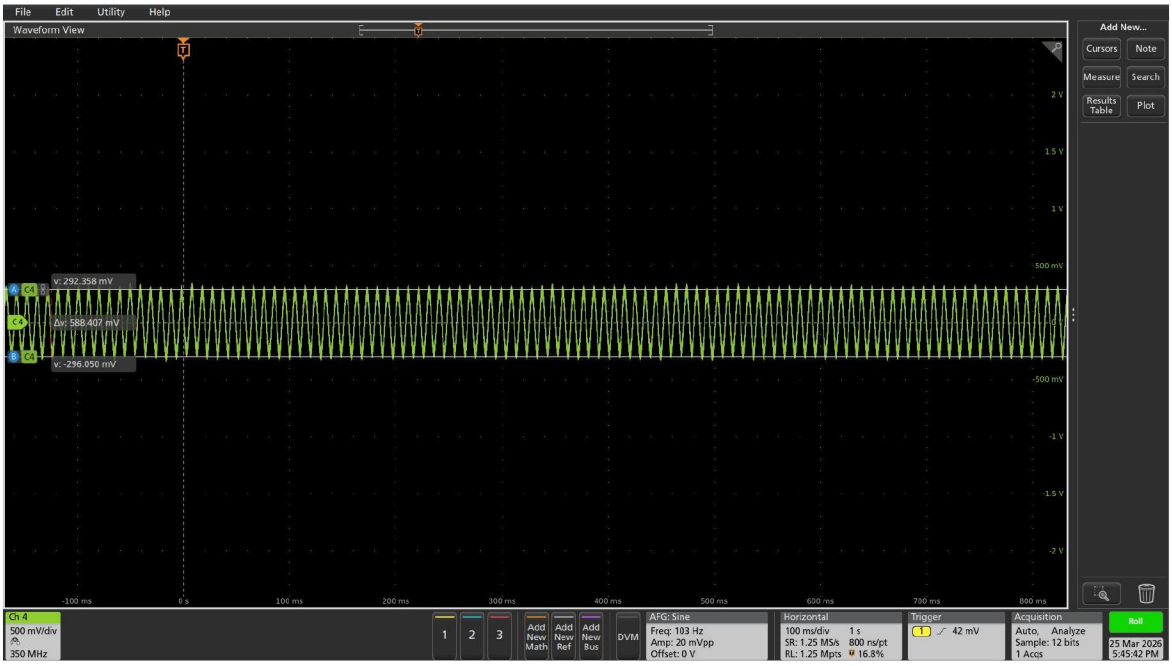


Fig. 5: Compensated op-amp near upper pole at $f = 103\,\text{Hz}$. The output is attenuated, confirming the dominant pole is below $100\,\text{Hz}$, consistent with the Bode plot showing gain rolloff beginning near 25–45 Hz. $\Delta V = 588.407\,\text{mV}$. Ch4: $500\,\text{mV/div}$.

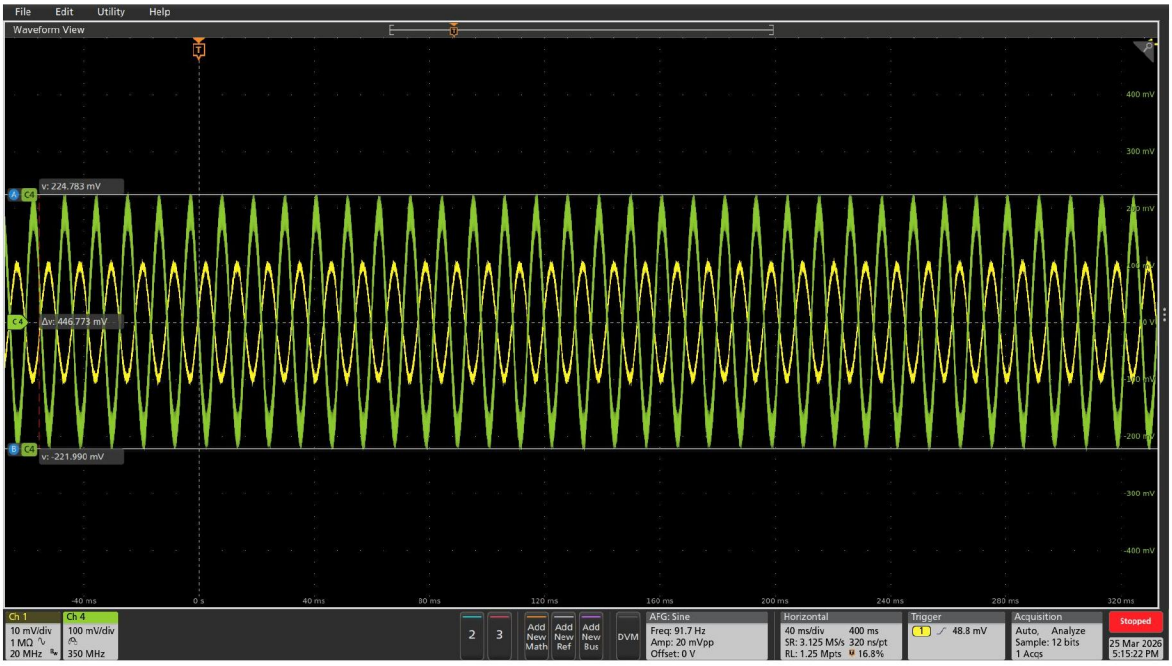
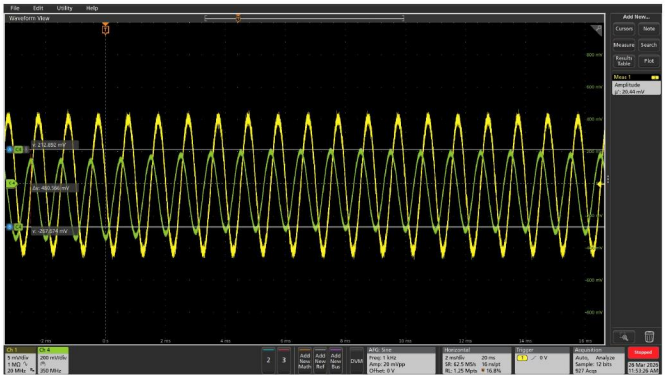
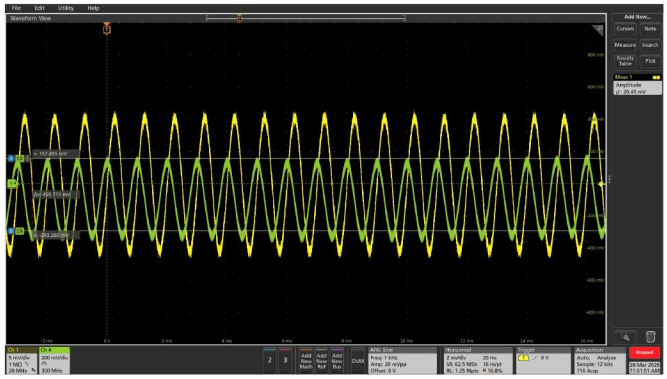


Fig. 6: Inverting amplifier midband at $f = 91.7\,\text{Hz}$. Yellow (Ch1, 10 mV/div): input; Purple (Ch4, 100 mV/div): output. $\Delta V_{\text{out}} = 446.773\,\text{mV}$; closed-loop gain $= 22.34$ V/V.

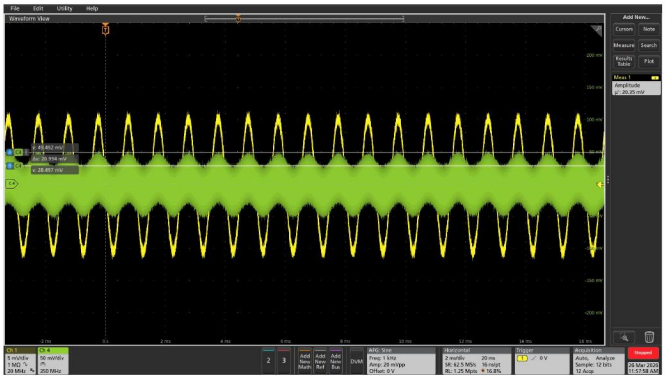


(a) Open-circuit output: $V_{OC} = 480.566\,\text{mV}$ at 1 kHz.

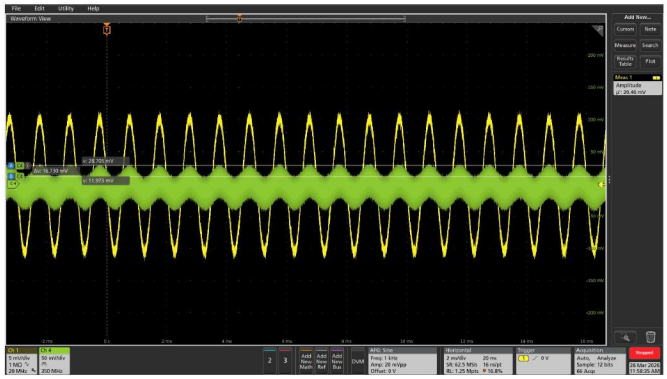


(b) Loaded output ($R_L = 196.2\,\Omega$): $V_L = 452.847\,\text{mV}$.

Fig. 7: Output impedance measurement at 1 kHz.



(a) Source voltage $V_S = 20.994\,\text{mV}$ with $R_S = 1958.4\,\Omega$.



(b) Input voltage at amplifier terminal: $V_{\text{in}} = 16.73\,\text{mV}$.

Fig. 8: Input impedance measurement at 1 kHz.

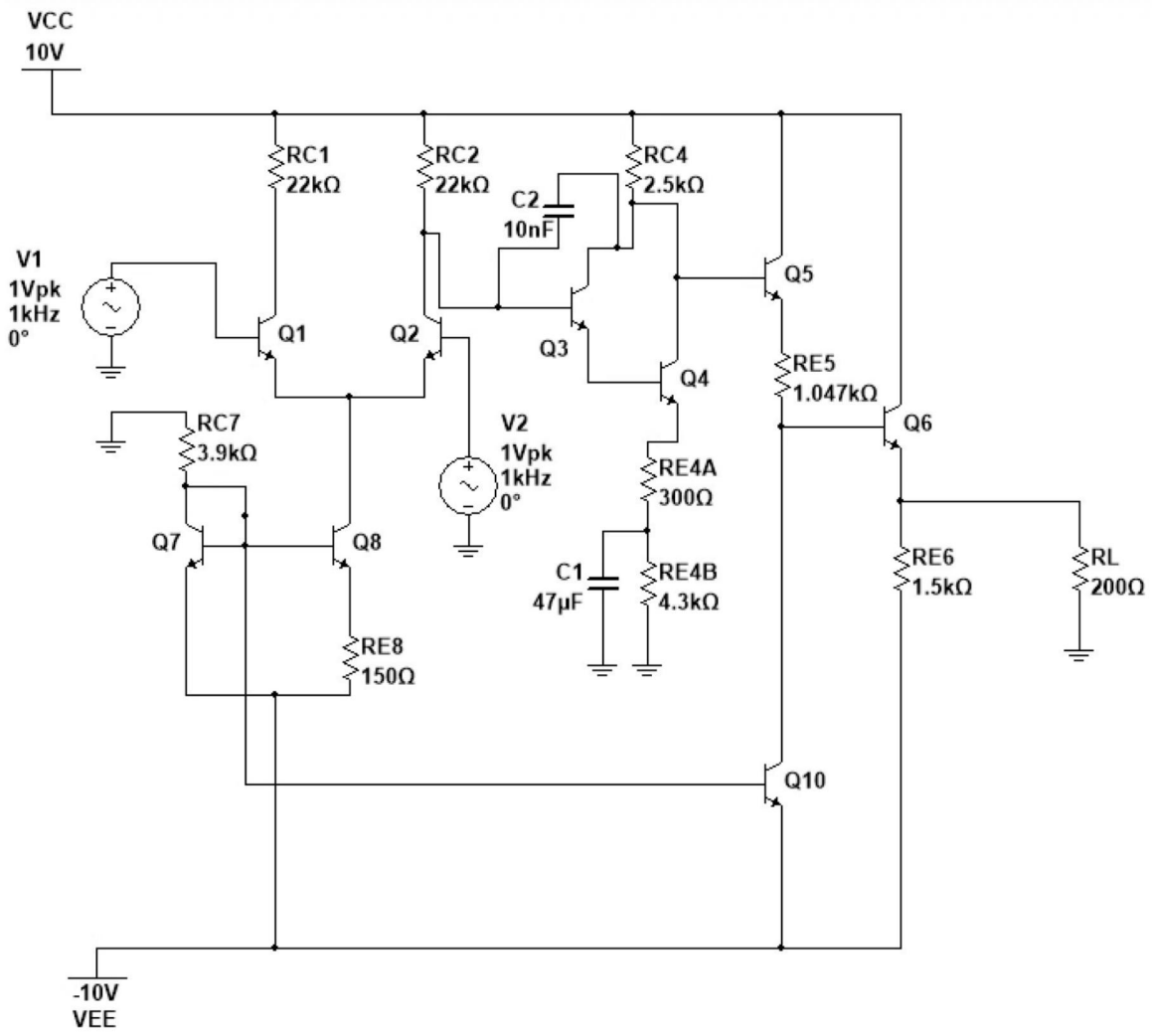


Fig. 9: Compensated op-amp schematic with $C_E = 47\,\mu\text{F}$ and $C_f = 10\,\text{nF}$. Key components: $R_{C2} = R_{C1} = 22\,\text{k}\Omega$, $R_{C4} = 2.5\,\text{k}\Omega$, $R_{E4A} = 300\,\Omega$, $R_{E4B} = 4.3\,\text{k}\Omega$.

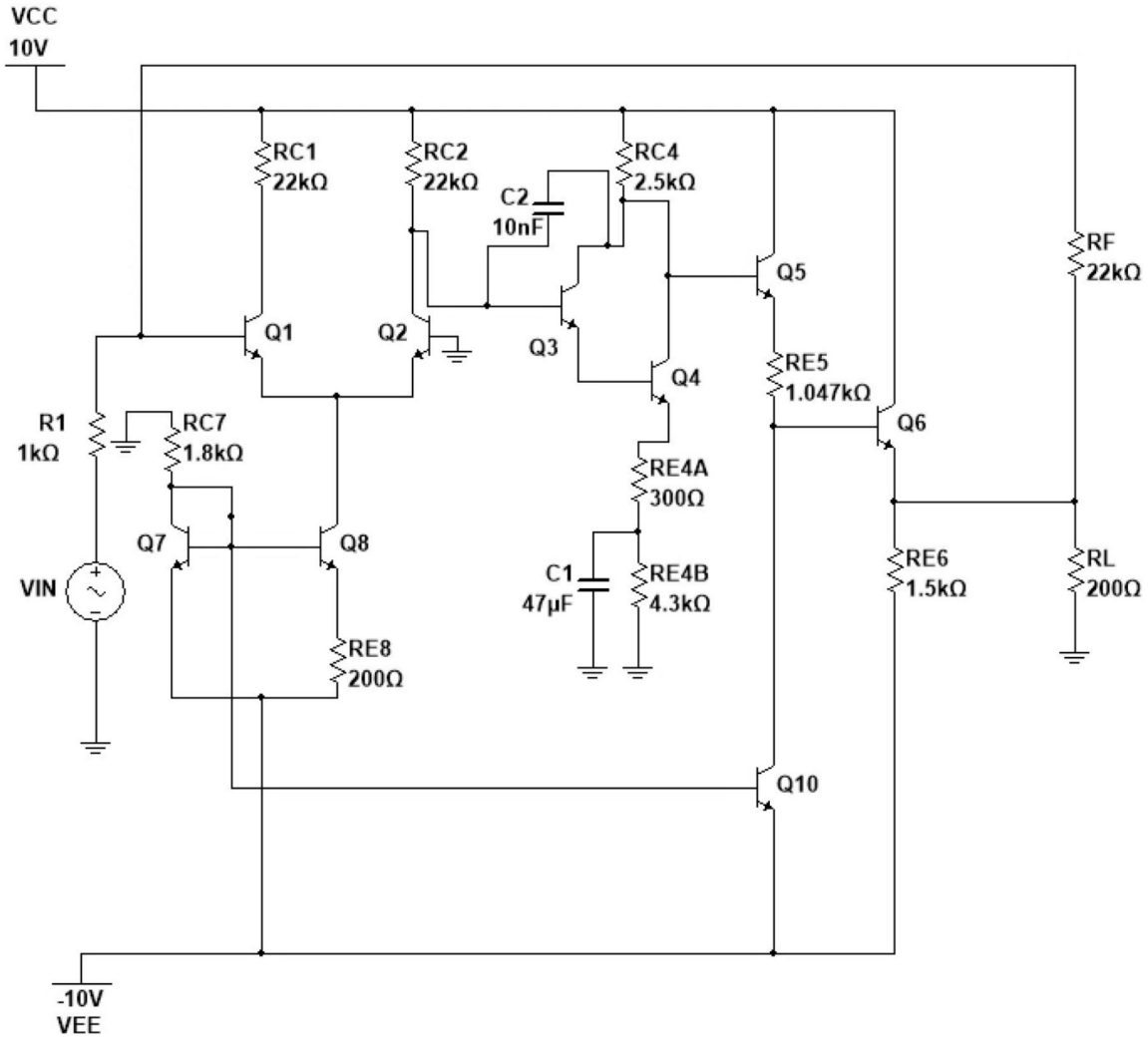


Fig. 10: Inverting amplifier schematic with $R_1 = 1\,\mathrm{k\Omega}$ and $R_f = 22\,\mathrm{k\Omega}$ connected to the compensated op-amp.